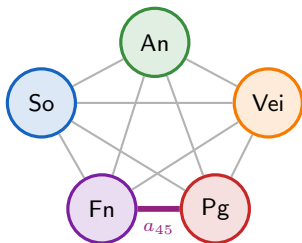


(a) Variational multiphase model



state at a material point:
volume fractions ϕ_i , viabilities ψ_i
 $\dot{\Psi} + \Delta + C \rightarrow \text{stat}$
 $\Psi = -\frac{1}{2}c^* \bar{\phi}^T \mathbf{A} \bar{\phi}, \bar{\phi}_i = \phi_i \psi_i$
 $\mathbf{A} = \mathbf{A}^T$: **15 unknowns θ**

(b) Data

5 species, Days 1–21
4 conditions:
commensal / dysbiotic
× static / HOBIC flow
composition + viability
pH: held out for checking

(c) Staged TMCMC on the GPU

